CDU-7500X

Conveyor Drive Unit

Operations & Maintenance Manual

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1. Safety Information

WARNING

Read all safety instructions before operating this equipment. Failure to follow safety procedures may result in serious injury or death. Never bypass, disable, or circumvent any safety interlock, guard, or emergency stop device.

1.1 Lockout/Tagout (LOTO) Procedure

Before performing any maintenance or repair work on the CDU-7500X:

1. Notify all affected personnel
2. Shut down the machine using the main control panel
3. Turn the main disconnect switch to OFF position
4. Apply your personal padlock to the disconnect switch
5. Attach a dated and signed lockout tag
6. Verify zero energy state by attempting to start the machine
7. Verify stored energy is dissipated (hydraulic, pneumatic, electrical)
8. Only then begin maintenance work

1.2 Personal Protective Equipment (PPE)

Required PPE when working on the CDU-7500X:

- Safety glasses with side shields
- Steel-toed safety boots
- Cut-resistant gloves (Class 3 minimum)
- Hearing protection (above 85 dB zones)
- High-visibility vest in active production areas

1.3 Emergency Stop

The CDU-7500X is equipped with three emergency stop buttons:

- Main control panel (red mushroom button)
- Drive end station
- Tail end station

To reset the emergency stop after activation:

1. Identify and resolve the cause of the emergency stop
2. Rotate the e-stop button clockwise to release
3. Press the RESET button on the control panel
4. Verify all guards are in place before restarting

2. Machine Specifications

2.1 General Specifications

Parameter	Value
Model	CDU-7500X
Conveyor Length	12,000 mm
Belt Width	800 mm
Belt Speed	0.3 m/s (adjustable 0.1-0.5 m/s)
Maximum Load	75 kg/m
Motor Power	7.5 kW
Motor Speed	1500 RPM
Supply Voltage	415V AC, 3-phase, 50 Hz
Operating Temperature	0 to 40 deg C ambient
Protection Rating	IP55
Weight (empty)	2,800 kg

2.2 Drive System

Parameter	Value
Drive Type	Helical gearbox with frequency inverter
Gear Ratio	25:1
Output Torque	450 Nm maximum
Drive Chain	ISO 10B-1, pitch 15.875 mm
Chain Tension	20 N +/- 3 N
Lubrication	ISO VG 68 mineral oil

2.3 Belt Specifications

Parameter	Value
Belt Type	EP 400/3 (3-ply polyester/nylon)
Belt Thickness	8 mm (3+2+3 cover)
Belt Tension (operating)	15 Nm +/- 2 Nm
Minimum Pulley Diameter	315 mm
Splice Type	Vulcanized finger splice

3. Installation

3.1 Foundation Requirements

The CDU-7500X requires a level concrete foundation with minimum compressive strength of 25 MPa. Foundation bolts: M16 x 200 mm, Grade 8.8, torque to 190 Nm.

3.2 Electrical Connection

1. Verify supply voltage: 415V AC +/-10%, 3-phase, 50 Hz
2. Cable size: minimum 4 sq mm copper, 4-core (3 phase + earth)
3. Circuit breaker: 32A Type D MCB
4. Earth resistance: less than 1 ohm
5. Motor insulation resistance: minimum 1 Megohm at 500V DC
6. Verify phase rotation before first start (clockwise viewing from fan end)

3.3 Belt Tracking Alignment

After installation, check belt tracking:
1. Run conveyor at low speed (0.1 m/s) with no load
2. Observe belt position at head and tail pulleys
3. If belt drifts to one side, adjust the corresponding idler
4. Turn adjustment screw maximum 1/4 turn at a time
5. Wait for 3 full belt revolutions between adjustments
6. Belt must run within 10 mm of center at all points
7. Lock adjustment screws after alignment is achieved

4. Belt Replacement Procedure

WARNING

Perform LOTO procedure (Section 1.1) before belt replacement. Belt replacement requires minimum 2 qualified technicians. Use appropriate lifting equipment for belt rolls (weight: approx 180 kg).

Step 1: Perform lockout/tagout as per Section 1.1
Step 2: Release belt tension by loosening the take-up screws (both sides equally)
Step 3: Disconnect the belt splice using the vulcanization cutter
Step 4: Remove the old belt by pulling from the tail end
Step 5: Clean all rollers, pulleys, and idlers; inspect for damage
Step 6: Thread the new belt over tail pulley, under return idlers, over drive pulley
Step 7: Prepare vulcanized finger splice per belt manufacturer specification
Step 8: Adjust belt tension to 15 Nm +/- 2 Nm (see Section 5)
Step 9: Perform belt tracking alignment (see Section 3.3)
Step 10: Run at 50% speed for 30 minutes, check tension and tracking
Step 11: Increase to full speed, run for 1 hour under observation
Step 12: Record belt replacement in maintenance log

Estimated time: 4-6 hours for experienced technicians.
Required tools: 17mm and 19mm wrenches, torque wrench (range 5-50 Nm), belt tension gauge, vulcanization press, belt cutter, lifting slings.

5. Belt Tension Adjustment

Proper belt tension is critical for reliable operation. Insufficient tension causes belt slip; excessive tension accelerates belt and bearing wear.

Target tension: 15 Nm +/- 2 Nm

Measurement procedure:
1. Stop the conveyor and perform LOTO
2. Place the tension gauge at the mid-span of the return side
3. Deflect the belt 25 mm and read the force
4. Calculate tension: T = F x L / 4 (where L = span in meters)
5. Acceptable range: 13 Nm to 17 Nm

Adjustment procedure:
1. Loosen the lock nuts on both take-up screws
2. Turn both take-up screws equally (maximum 1/2 turn at a time)
3. Clockwise to increase tension, counter-clockwise to decrease
4. Re-measure tension after each adjustment
5. Tighten lock nuts to 25 Nm when target is achieved
6. Verify belt tracking after tension adjustment

6. Error Codes

The CDU-7500X control system displays error codes on the HMI panel. All error codes are prefixed with 'E-' followed by a three-digit number.

6.1 Error Code Reference Table

Code	Description	Action Required
E-100	Drive overload	Check belt tension, load, and motor current
E-101	Drive overcurrent	Check motor windings and connections
E-150	Inverter fault	Reset inverter; check for short circuit
E-200	Belt slip detected	Check belt tension (Section 5)
E-201	Belt misalignment	Check belt tracking (Section 3.3)
E-247	Motor overheat	Check ventilation duct, clean fan guard, verify ambient temp below 40 deg C. If E-247 persists after cleaning, check thermal relay setting (95 deg C +/- 5 deg C). Inspect motor bearings for excessive friction.
E-300	Emergency stop active	Identify cause, reset e-stop (Section 1.3)
E-301	Safety guard open	Close all safety guards before operation
E-400	Communication fault	Check PLC-HMI cable, restart controller
E-500	Encoder failure	Check encoder cable and connections

7. Lubrication Schedule

Use only approved lubricants. Mixing different lubricant types may cause equipment damage.

Parameter	Value
Drive chain	ISO VG 68 mineral oil, every 500 hours
Gearbox	ISO VG 220 synthetic, check monthly, change every 5000 hours
Drive bearings	NLGI Grade 2 lithium grease, every 2000 hours
Idler bearings	NLGI Grade 2 lithium grease, every 4000 hours
Take-up screws	Anti-seize compound, every belt change

7.1 Lubrication Points

The CDU-7500X has 12 lubrication points:
- 2x drive shaft bearings (grease nipples, left and right)
- 2x tail shaft bearings (grease nipples, left and right)
- 4x return idler bearings (grease nipples)
- 1x drive chain (drip oiler or manual application)
- 1x gearbox (oil level sight glass, drain plug at bottom)
- 2x take-up screw threads (manual application)

All lubrication points are marked with yellow tags on the machine.

8. Maintenance Schedule

8.1 Daily Checks

- Visual inspection of belt condition and tracking
- Check for unusual noise or vibration
- Verify all safety guards are in place
- Check drive chain tension (visual)
- Record operating hours

8.2 Weekly Checks

- Measure belt tension (target: 15 Nm +/- 2 Nm)
- Inspect belt edges for wear or damage
- Check motor current (should be below 80% of rated)
- Clean photoelectric sensors
- Check emergency stop function

8.3 Monthly Checks

- Gearbox oil level (top up if below sight glass)
- Drive chain tension measurement
- Roller and idler rotation check
- Electrical connection tightness
- Motor insulation resistance (minimum 1 Megohm)

8.4 Annual Overhaul

- Complete belt inspection and replacement if worn
- Gearbox oil change
- All bearing lubrication
- Electrical system inspection
- Safety device testing and certification
- Alignment verification

9. Troubleshooting Guide

Symptom	Possible Cause	Corrective Action
Belt slipping	Low belt tension	Adjust tension to 15 Nm (Section 5)
Belt slipping	Wet or oily belt surface	Clean belt and pulleys; identify source of contamination
Belt drifting to one side	Misaligned idlers	Realign idlers (Section 3.3)
Belt drifting to one side	Uneven loading	Check material feed distribution
Excessive vibration	Bearing failure	Replace bearings; check alignment
Excessive vibration	Damaged roller	Replace damaged roller
Motor overheating (E-247)	Blocked ventilation	Clean motor fan and ventilation duct
Motor overheating (E-247)	Overload condition	Check load; verify belt tension
Motor overheating (E-247)	High ambient temperature	Ensure ambient below 40 deg C
Unusual noise from drive	Worn drive chain	Inspect and replace chain if elongation > 2%
Unusual noise from drive	Low gearbox oil	Check and top up gearbox oil level
Conveyor won't start	E-stop active	Reset e-stop (Section 1.3)
Conveyor won't start	Safety guard open	Close all guards; check interlock switches

10. Emergency Procedures

WARNING

In case of fire, injury, or equipment failure that poses immediate danger: 1) Press the nearest emergency stop button. 2) Evacuate the area. 3) Call emergency services. 4) Notify your supervisor. Do NOT attempt to restart equipment after a serious incident.

10.1 Belt Jam Procedure

1. Press emergency stop immediately
2. Do NOT attempt to clear the jam while any part is moving
3. Perform LOTO (Section 1.1)
4. Clear the jammed material manually
5. Inspect belt for damage before restarting
6. Document the incident in the maintenance log

10.2 Electrical Emergency

1. Do NOT touch the equipment if electrical fault is suspected
2. Turn off main disconnect from a safe distance using insulated tools
3. If a person is in contact with live equipment, do NOT touch them
4. Call emergency services immediately
5. Use appropriate fire extinguisher (CO2 or dry powder) for electrical fires